

Graphs

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Solution of the practice quiz

The incidence function is defined as

$$\begin{aligned}\Phi(e_1) &= (v_0, v_2), & \Phi(e_2) &= (v_0, v_3), & \Phi(e_3) &= (v_2, v_0), \\ \Phi(e_4) &= (v_3, v_4), & \Phi(e_5) &= (v_5, v_3), & \Phi(e_6) &= (v_1, v_4), \\ \Phi(e_7) &= (v_1, v_6), & \Phi(e_8) &= (v_4, v_7), & \Phi(e_9) &= (v_5, v_7), \\ \Phi(e_{10}) &= (v_2, v_1).\end{aligned}$$

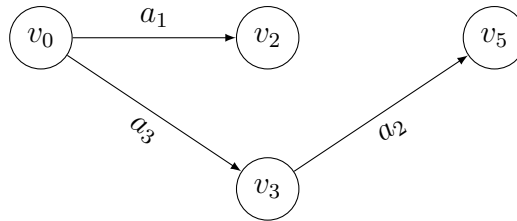
The incidence function is injective, as there is at most one arc linking any pair of nodes.

Recall that given a graph $(\mathcal{V}, \mathcal{E}, \Phi)$, we define $(\mathcal{V}', \mathcal{E}', \Phi')$ as a subgraph if

- $\mathcal{V}' \subseteq \mathcal{V}$,
- $\mathcal{E}' \subseteq \mathcal{E}$,
- $\Phi'(e) = \Phi(e)$, for each $e \in \mathcal{E}'$,
- for each $e \in \mathcal{E}'$, if $\Phi'(e) = (i, j)$, then i and j belong to $\mathcal{V}'$.

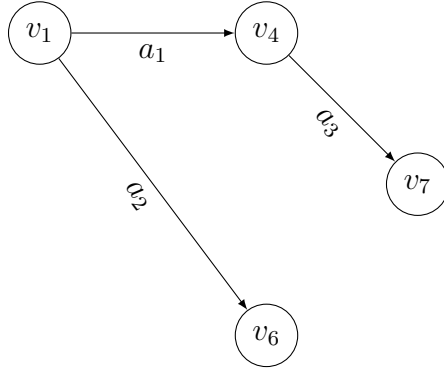
We analyze now the three cases.

1. First graph.



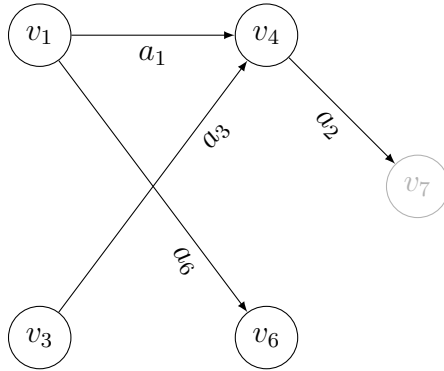
We have $\mathcal{V}_1 \subseteq \mathcal{V}$. Moreover, it can easily be verified that for each $a \in \mathcal{E}_1$, if $\Phi_1(a) = (i, j)$, then i and j belong to $\mathcal{V}_1$. As we need $\Phi_1(a_1) = \Phi(a_1) = (v_0, v_2)$, we define $a_1 = e_1$. Similarly, we define $a_3 = e_2$. Therefore, both a_1 and a_3 belong to $\mathcal{E}$. But we cannot find a corresponding arc for a_2 in the original graph, as there is no arc linking nodes v_3 and v_5 . Therefore, $\mathcal{E}_1 \not\subseteq \mathcal{E}$, and $(\mathcal{V}_1, \mathcal{E}_1, \Phi_1)$ is not a subgraph of $\mathcal{G}$.

2. Second graph.



We have $\mathcal{V}_1 \subseteq \mathcal{V}$. Moreover, it can easily be verified that for each $a \in \mathcal{E}_2$, if $\Phi_2(a) = (i, j)$, then i and j belong to $\mathcal{V}_2$. As we need $\Phi_2(a_1) = \Phi(a_1) = (v_1, v_4)$, we define $a_1 = e_6$. Similarly, we define $a_2 = e_7$ so that $\Phi_1(a_2) = \Phi(a_2) = \Phi(e_7) = (v_1, v_6)$. Finally, we define $a_3 = e_8$ so that $\Phi_1(a_3) = \Phi(a_3) = \Phi(e_8) = (v_4, v_7)$. Therefore, all arcs in $\mathcal{E}_2$ belong to $\mathcal{E}$, the two incidence functions are consistent, and $(\mathcal{V}_2, \mathcal{E}_2, \Phi_2)$ is a subgraph of $\mathcal{G}$.

3. Third graph.



We have $\mathcal{V}_3 \subseteq \mathcal{V}$ and $\mathcal{E}_3 \subseteq \mathcal{E}$. The arcs are also incident to the same nodes as in the original graph, so that the third property is verified. But the last property is not satisfied: $(v_4, v_7) \in \mathcal{E}_3$ but $v_7 \notin \mathcal{V}_3$. Thus, $(\mathcal{V}_3, \mathcal{E}_3, \Phi_3)$ is not a subgraph of $\mathcal{G}$.